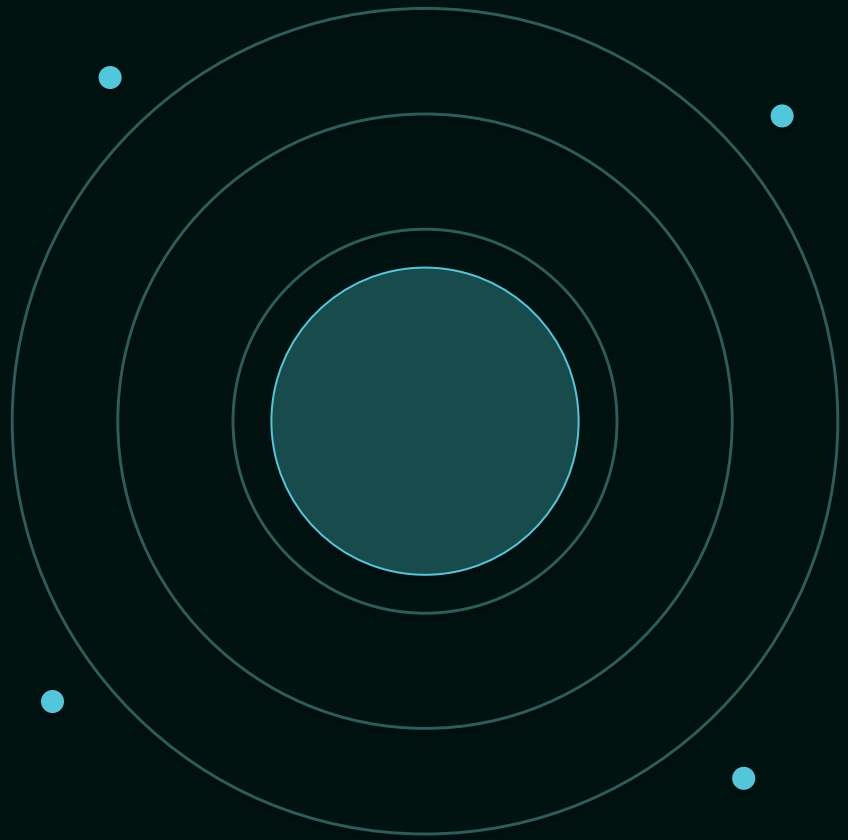
2026 NASA SPACE APPS CHALLENGE

INTERIM

ASTRONAUT HEALTH ASSESSMENT & DECISION-MAKING SIMULATOR

Making health decisions when communication with Earth is delayed.



TEAM • A++

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LIVE DEMO | https://limlining.github.io/Nasa-Interim/

GITHUB | github.com/limlining/Nasa-Interim

Long-duration missions create health-monitoring challenges

Astronauts face spaceflight hazards that can affect physical and psychological health.

NASA'S FIVE HAZARDS OF HUMAN SPACEFLIGHT

01 Space radiation

02 Isolation and confinement

03 Distance from Earth

04 Gravity fields

05 Hostile / closed environments

POTENTIAL HEALTH IMPACTS

- Immune system changes
- Cardiovascular risks
- Bone loss
- Sleep disturbances
- Behavioural health concerns

Astronauts need to recognise changes in their health.

Challenge question: How can astronauts monitor health indicators, evaluate their health status and take appropriate action?

How INTERIM addresses the challenge

The current prototype focuses on health awareness, assessment practice and response decisions.

01 GATHER / VIEW INDICATORS

Health Indicators

Displays simulated vital signs such as heart rate, SpO_2 and blood pressure, alongside changing scenario information.

02 EVALUATE

Health Assessment

Users review symptoms and indicator trends to assess the simulated astronaut's condition and urgency.

03 ACT

Decision & Action

Users choose from four response options and follow an action checklist within the scenario.

CURRENT SCOPE

Health indicators and recommendations are simulated. Real-time wearable sensor integration is not implemented.

INTERIM adds communication-delay and blackout scenarios to practise decisions when ground support is limited.

What is INTERIM?

A web-based educational prototype that lets users experience astronaut health scenarios and practise decisions under communication constraints.

01

EXPERIENCE

Enter an astronaut health scenario

→

02

ASSESS

Review the symptoms and situation

→

03

DECIDE

Choose a response within the time limit

→

04

REFLECT

Compare with simulated ground advice

INTERIM supports awareness and decision-making practice — it does not provide medical diagnosis.

Six simulated astronaut health situations

Users select a scenario, review the situation and decide how to respond.

S1 **CHEST PAIN**

Chest pain and elevated heart rate

S2 **INSOMNIA & LOW MOOD**

Sleep problems and mood changes

S3 **ELEVATED CO_2**

High CO_2 levels with headache

S4 **LOW BLOOD PRESSURE**

Dizziness and low blood pressure

S5 **POST-EVA SYMPTOMS**

Nausea and skin redness after EVA

S6 **IRREGULAR HEARTBEAT**

Palpitations and irregular heartbeat

What users can do in INTERIM

The following capabilities are implemented in the current prototype.

01

Six health scenarios

Chest pain • insomnia / low mood • elevated CO_2 • low blood pressure • post-EVA symptoms • irregular heartbeat

02

Communication modes

Toggle between delayed ground communication and a complete communication blackout.

03

Live simulation

Watch simulated indicators, trends, countdown and a “body budget” change as time passes.

04

Action checklist

Choose Observe, Self-treat, Call crew or Emergency; follow a step-by-step checklist.

05

Human-body navigation

Explore scenario entry points through an interactive body-region interface.

06

Progress review

Daily wake-up check-in, saved Decision Log, and a Performance Profile after 3 unaided scenarios.

From check-in to reflection

A repeatable flow supports practice before, during and after a simulated event.

01
CHECK IN
Sleep & readiness

02
SELECT
Scenario or body region

03
ASSESS
Read symptoms & indicators

04
DECIDE
Choose one of 4 actions

05
WAIT / ACT
Delay or blackout simulation

06
REVIEW
Compare, log & reflect

DECISION LOOP

Observe → choose → monitor → escalate if needed → compare with simulated guidance → learn

AFTER A SCENARIO

The user receives simulated feedback, then can review completed scenarios in the Decision Log and Performance Profile.

One scenario. Four possible responses.

Example: chest pain with elevated heart rate — the system simulates changing indicators while the user decides.

SCENARIO S1

CHEST PAIN + HEART RATE 112

Crushing chest pressure spreads to the left arm after exercise.

Simulated ground delay: 8 min

SIMULATED INDICATORS

HEART RATE	SpO_2
112 bpm	95%

BLOOD PRESSURE	CHEST PAIN
148 / 92 mmHg	6 / 10

ACTION CHOICES

1. Observe
2. Self-treat
3. Call crew
4. Emergency

The prototype compares the selected action with a simulated reference response. Values and recommendations are illustrative — not clinical thresholds.

What INTERIM demonstrates — and what comes next

A transparent prototype distinguishes its current capabilities from future development.

INTENDED IMPACT

Practice & awareness

- Builds awareness of spaceflight health risks
- Practises assessment of simulated indicators
- Encourages timely decisions and crew escalation
- Supports reflection through saved results

CURRENT LIMITATIONS

Prototype boundaries

- Uses simulated health data and ground advice
- No real-time sensor integration
- No clinically validated decision rules
- Not for diagnosis or treatment

FUTURE DEVELOPMENT

Research-informed next steps

- Explore relevant NASA health datasets
- Integrate approved wearable devices
- Add physiological and psychological indicators
- Improve trend analysis and reporting
- Validate scenarios with medical experts

Design principle: support crew judgement — never replace the astronaut or medical team.

Recognise. Assess. Decide. Reflect.

INTERIM helps users practise evaluating simulated astronaut health situations and choosing a response when immediate ground support is limited.

INTERIM is an educational simulation prototype — not a real-time medical monitoring system.

LIVE DEMO https://limlining.github.io/Nasa-Interim/

GITHUB | github.com/limlining/Nasa-Interim

Thank you

OFFICIAL REFERENCES

NASA Space Apps Challenge

Create Health Monitoring Software for Astronauts on Space Missions

NASA Human Research Program

5 Hazards of Human Spaceflight

NASA Human Research Program

Risks of Human Spaceflight

NASA Human Research Program

Ongoing HRP Research

Project status: simulated educational prototype. Not an official NASA product.